GaugeFlow: Equivariance, Not Invariance, Is the Right Prior for Continuous Physical Gauges in Self-Supervised Medical Imaging

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Abstract

A self-supervised objective that works on one medical-imaging task often collapses on the next. We give a mechanistic account of when it transfers and when it does not, organized around a single distinction: whether a task's nuisance axis is a *continuous physical gauge* with *coincident free supervision*, and whether the downstream utility is invariant or covariant under that gauge. Contrast-front propagation in digital subtraction angiography (DIAS) is such a gauge: it is a real temporal dynamical process, so "predict the next state" is gauge-equivariance under time-translation with free supervision, and a compact gauge-consistent objective improves DIAS adjacent-frame prediction while keeping topology-style leakage non-positive. We then ask whether the same prior transfers to a second continuous physical gauge, the projection angle of coronary angiography (CardioSYNTAX), where same-study/same-artery retrieval is the utility. Adversarial invariance via gradient reversal—the standard tool for suppressing a nuisance—is a no-free-lunch wall here: across a swept adversary strength, no setting holds retrieval non-inferior while meeting the angle-leakage cap. We test the principled alternative, gauge-equivariance, by predicting how the embedding should transform under a known Δangle through an $SO(2)$ head. As an auxiliary it strictly dominates the adversary, removing its statistically significant harm (-0.044, $p \approx 0 \rightarrow -0.018$, $p = 0.11$) and demonstrably using the gauge (real Δangle 0.079 vs. shuffled 0.028). As the *primary* objective it fails, and the failure is diagnostic: same-study-across-angles retrieval is itself an invariance metric, and on this encoder the projection-angle gauge is not a recoverable group action—transport and shuffled-transport are indistinguishable (0.0093 vs. 0.0089) under both common-reference canonicalization and the rigorous pairwise-transport evaluation. The contribution is a predictive rule: gauge-equivariant self-supervision helps exactly when the utility is gauge-*covariant* and the gauge acts as a recoverable transformation of the representation; where the utility is invariance-shaped, no amount of gauge machinery beats simply not fighting the gauge. All code and the full per-seed evidence are released.

1 Introduction

Self-supervised and nuisance-suppression objectives in medical imaging are usually justified by an intuition—"factor out the scanner," "be invariant to acquisition"—and then validated on one benchmark. The uncomfortable empirical fact is that the same objective frequently helps on the task it was tuned for and silently hurts elsewhere. This paper replaces the intuition with a mechanism. We claim that the deciding property is not the modality or the architecture but a pair of structural facts about the task: (i) whether the nuisance axis is a *continuous physical gauge* that comes with coincident, free supervision, and (ii) whether the downstream utility is *invariant* or *covariant* under that gauge. When both align—a continuous gauge with a covariant target—a gauge-aware objective is principled and helps. When they do not, the standard move of suppressing the gauge with an adversary is not merely suboptimal; it is provably caught on an accuracy-versus-invariance frontier with no free lunch.

We arrive at this through angiography. Contrast-front propagation in DIAS is a genuine temporal dynamical process: paired frames differ by the gauge (contrast phase) while sharing the same underlying anatomy-state transition, and "predict the future observation" is therefore gauge-equivariance under time-translation, with the next frame supplying free supervision. A compact gauge-consistent trainer improves DIAS adjacent-frame prediction over a fixed persistence comparator while holding topology-style leakage non-positive (Section 4). The interesting science is what happens on a *second* continuous physical gauge that lacks the temporal coincidence: the projection angle of an X-ray angiography acquisition.

On that gauge we make three findings. First, adversarial invariance via gradient reversal [Ganin et al., 2016]—the reflexive tool for removing a nuisance—is a no-free-lunch wall: sweeping its strength traces a monotonic frontier on which no operating point simultaneously keeps retrieval non-inferior and drives angle-leakage under cap (Section 5). Second, the principled alternative suggested by the DIAS success—gauge-*equivariance*, i.e. predicting how the representation should transform under a known gauge action rather than throwing the gauge away—strictly dominates the adversary when used as an auxiliary, eliminating its significant harm and provably exploiting the gauge signal (Section 6). Third, promoted to the primary objective, equivariance fails, and the failure is the point: same-study-across-angles retrieval is structurally an invariance metric, and the projection-angle gauge is not a recoverable group action on the encoder (Section 7). We confirm the third finding is protocol-independent with a rigorous pairwise-transport evaluation, not merely an artifact of how we canonicalize.

The result is a usable rule rather than another benchmark number (Section 8): gauge-equivariant self-supervision helps when, and only when, the task utility is gauge-covariant and the gauge acts as a recoverable transformation of the representation. This explains both the DIAS positive and the angiography boundary with one mechanism, and it tells a practitioner—before training—whether a gauge-aware objective can possibly help on a new task.

2 The Gauge Frame

Gauges, invariance, equivariance. Let a *gauge* be a controlled, label-preserving change of observation: a transformation g of the input that leaves the anatomy-state label fixed while changing the appearance. A representation $z = f(x)$ is gauge-*invariant* if $f(g \cdot x) = f(x)$ and gauge-*equivariant* if there is a known action ρ_g on representation space with $f(g \cdot x) = \rho_g f(x)$. Invariance discards the gauge; equivariance keeps it in a structured, predictable form. The two objectives sit on opposite sides of an information trade: invariance is information-destroying by construction, equivariance is information-preserving.

Continuous physical gauges with coincident supervision. A gauge is a *continuous physical gauge* when it is a real, measurable, continuously parameterized degree of freedom of the acquisition—contrast phase over time, projection angle in degrees—rather than a discrete derived label (a style cluster, a vendor bucket). It carries *coincident supervision* when the gauge parameter is observed for free alongside the data and, in the strongest case, when moving along the gauge is itself the prediction target. DIAS contrast propagation is the strong case: the gauge parameter (time) advances and the next frame is the label, so equivariance under the gauge *is* the task. Projection angle is a continuous physical gauge with the parameter observed (the positioner angle is recorded) but without temporal coincidence: the angle is known, yet no free target says "this is the same anatomy seen $\Delta\theta$ later."

The covariance condition. The deciding question for whether a gauge-aware objective can help is whether the downstream utility is invariant or covariant under the gauge. If the utility rewards treating gauge-separated views as the same—e.g. retrieving the same study across acquisition angles—then the metric is itself an invariance objective, and keeping the gauge information works against it. If the utility depends on the gauge—e.g. predicting a gauge-dependent future state—then discarding the gauge destroys signal the task needs, and equivariance is the principled prior. This single condition, applied before training, predicts the outcomes in this paper.

The objective. For a paired observation (x_a, x_b) of the same cluster differing by gauge value (θ_a, θ_b), with encoder $z = f(x)$, we compare three training signals. *Adversarial invariance* adds a gradient-reversal head that predicts θ_a from z_a and is trained to fail, pushing the gauge out of z. *Equivariance (auxiliary)* keeps the base task and adds a head h that predicts the paired embedding from the gauge displacement, $\mathcal{L}_{\text{eq}} = \|h(z_a, \Delta\theta) - \text{sg}(z_b)\|^2$, with $\Delta\theta = \theta_b - \theta_a$ encoded as $(\sin\Delta\theta, \cos\Delta\theta)$ to respect the $SO(2)$ structure of a real angle, and a stop-gradient on the target so the contrastive negatives prevent collapse. *Equivariance (primary)* routes the contrastive positive itself through the head—transporting z_a into z_b's gauge frame, $h(z_a, \Delta\theta)$, then contrasting against z_b in both directions—so the only way to satisfy the task is for the gauge to act as a predictable transformation; at evaluation, views are compared after transport into a common gauge frame.

3 Experimental Contract

Every external number is the output of one pre-registered analyzer: cluster-bootstrap confidence intervals, a permutation null on the paired per-cluster delta, and a shuffled-gauge negative control that repartners views at random so the gauge displacement becomes meaningless. A method passes a gauge only if it holds the task non-inferior to baseline, meets the gauge-leakage cap, and the shuffled-gauge control reads as null. One portable trainer is used throughout; variants differ only by configuration flags, so the no-op configuration reproduces the baseline exactly.

The projection-angle gauge uses CardioSYNTAX coronary angiography on real local shards across five seeds, roughly 200 study clusters per arm. A precondition for an honest equivariance test is that the gauge is genuinely continuous; we verified this before training. The recorded primary positioner angle takes 629 distinct values over $[-46.4°, 47.5°]$ (standard deviation 19.8°), and within-study angle displacements have median 21.7° with only 5.4% exactly zero—so the 11 coarse buckets used as a categorical gauge label conceal a real continuous signal that an $SO(2)$ head can exploit. The task is same-study/same-artery retrieval@1 (higher is better); the leakage probe is the fold-fit R^2 with which the angle can be recovered from the representation (cap 0.035, the raw-feature reference). The contrast-phase gauge uses the DIAS curated-v1 benchmark against the confirmed persistence comparator.

4 DIAS: The Gauge That Works

On the contrast-phase gauge—the one continuous physical gauge in this study whose utility is covariant, because the task is to predict the gauge-advanced future frame—the gauge-consistent objective helps. The balanced GaugeFlow variant improves all accepted persistence-baseline metrics: next-frame test MAE 0.01142 versus persistence 0.01153, with a topology-style leakage probe delta of -0.074 (non-positive, the leakage gate) and bounded commutator and state-consistency diagnostics (Table 1). This is the expected outcome under the gauge frame: time advancement supplies free supervision, the target is gauge-covariant, and equivariance under time-translation is exactly what the benchmark rewards.

Table 1: DIAS contrast-phase gauge (covariant utility). The balanced variant is the first tested point that clears both the prediction and the topology-style leakage gates.

Variant	Next-frame MAE	Topology-style leakage Δ	Gate
Persistence comparator	0.011530	—	—
GaugeFlow initial	0.011518	$+0.021$	leakage positive, fail
GaugeFlow balanced	0.011418	-0.074	both gates clear

5 Adversarial Invariance Is a No-Free-Lunch Wall

Moving to the projection-angle gauge, the reflexive way to suppress the nuisance is a gradient-reversal adversary on the embedding. It does not work, and not because it is mistuned. A single-embedding adversary is actively harmful: retrieval falls from 0.0821 to 0.0466, a delta of -0.044 (CI $[-0.063, -0.025]$, permutation $p \approx 0$), while angle-leakage stays near baseline at $R^2 = 0.31$ against a 0.035 cap. Porting the full content/style disentanglement of the DIAS trainer—a dedicated gauge-absorbing path—repairs the destructiveness, cutting angle-leakage roughly threefold ($0.27 \to 0.09$) and shrinking the harm to -0.014, but it still does not clear the cap while holding retrieval non-inferior.

Sweeping the adversary strength shows this is a frontier, not a tuning miss (Table 2). With the adversary off, the consistency-only objective is non-inferior ($+0.0055$, $p = 0.55$) but leaves leakage at 0.265; any positive adversary weight drives leakage down at a monotone cost to retrieval. No swept setting simultaneously holds retrieval non-inferior and meets the leakage cap. This is the classic invariance information trade made concrete: forcing the gauge out of a representation that the retrieval task wants to keep coherent necessarily spends task accuracy.

Table 2: Projection-angle gauge, adversary-strength sweep. The frontier is monotone: invariance pressure trades retrieval for leakage with no free lunch.

Adversary weight	Retrieval Δ vs. baseline [CI], p	Angle-leakage R^2
0.00 (consistency only)	$+0.0055$ $[-0.012, +0.024]$, $p = 0.55$	0.265
0.05 (weak)	-0.044, harmful	lower
0.25 (full)	-0.044 $[-0.063, -0.025]$, $p \approx 0$	0.31

6 Equivariance Dominates Invariance as an Auxiliary

The gauge frame says the principled move is to keep the gauge in a structured form rather than fight it. We add an $SO(2)$ equivariance head that predicts the paired embedding from the angle displacement, leaving the retrieval task intact. It strictly dominates the adversary. Retrieval delta improves from the adversary's significant -0.044 ($p \approx 0$) to -0.018 (CI $[-0.040, +0.004]$, $p = 0.11$): the harm is halved and is no longer statistically distinguishable from zero. Crucially, the head provably uses the gauge—with the true angle displacement, retrieval is 0.079, but with the shuffled-gauge control it drops to 0.028, a gap of 0.051 that the adversary has no analogue for. Leakage stays high (0.30) by design, because equivariance keeps the information rather than destroying it. As an auxiliary, then, equivariance buys back the accuracy the adversary spends and reads the gauge signal the adversary erases (Table 3).

7 Equivariance as Primary Objective: A Diagnostic Failure

Promoted from auxiliary to primary objective—routing the contrastive positive through the transport head and retrieving after canonicalization to a common gauge frame—equivariance fails sharply: retrieval collapses to 0.0089 against a baseline of 0.095, a delta of -0.080 ($p \approx 0$), with leakage shooting to 0.70. The failure is informative because the gauge signal vanishes: the true angle displacement (0.0089) is now indistinguishable from the shuffled control (0.0105), the opposite of the auxiliary regime.

Two structural facts explain this. First, same-study-across-angles retrieval treats different angles of the same study as positives, so the metric *is* an invariance objective; an equivariant representation that faithfully preserves the angle separates exactly the views the metric wants together. Second, the projection-angle gauge does not act as a recoverable group action on this encoder: the transport head cannot map an embedding from one angle frame to another in a way that recovers study identity, so canonicalization scrambles rather than aligns.

We rule out the obvious objection—that common-reference canonicalization is simply a bad retrieval protocol—with the rigorous evaluation an equivariant representation deserves: pairwise transport, moving each query into every candidate's gauge frame before scoring. It gives the same answer. Retrieval is 0.0093 for the true gauge versus 0.0089 for the shuffled control, a null gap, with leakage 0.73 (Table 3). The negative is therefore protocol-independent: transport and shuffled-transport are indistinguishable however we evaluate, which is direct evidence that the gauge is not a recoverable transformation of the representation here.

Table 3: The invariance–equivariance frontier on the projection-angle gauge. Equivariance dominates adversarial invariance as an auxiliary and uses the gauge; as a primary objective it fails because the retrieval utility is invariance-shaped and the gauge is not a recoverable group action. Per-run baselines differ by $\sim$0.014 from MPS nondeterminism, so the valid statistic is the within-run paired delta.

Objective	Retrieval Δ vs. baseline	Gauge used? (true vs. shuffled)	Leakage R^2
Adversarial invariance (single)	-0.044, $p \approx 0$	—	0.31
Adversarial invariance (dual path)	-0.014	—	0.09
Equivariance, auxiliary	-0.018, $p = 0.11$ (n.s.)	0.079 vs. 0.028 (used)	0.30
Equivariance, primary (canonical)	-0.080, $p \approx 0$	0.0089 vs. 0.0105 (none)	0.70
Equivariance, primary (pairwise)	-0.050, $p \approx 0$	0.0093 vs. 0.0089 (none)	0.73

8 When Do Physical Gauges Help?

The two gauges resolve into one rule. DIAS is a continuous physical gauge with coincident supervision *and* covariant utility—the task is to predict the gauge-advanced state—so equivariance under the gauge is the task and the gauge-consistent objective helps. Projection angle is an equally continuous, equally physical gauge, but its retrieval utility is invariant, and the gauge does not act as a recoverable transformation of the encoder; there, keeping the gauge fights the metric and discarding it via an adversary spends accuracy on a no-free-lunch frontier. The best available move on an invariance-shaped task is the cheapest one—do not fight the gauge at all—which is exactly why the consistency-only and auxiliary-equivariance settings, both of which leave the task intact, land at parity while every aggressive gauge objective underperforms.

This yields a pre-training decision rule. Gauge-equivariant self-supervision is the right prior when: the nuisance is a continuous physical gauge with an observed parameter; the downstream utility is *covariant* under that gauge; and the gauge acts as a recoverable transformation of the representation. When the utility is invariance-shaped, no gauge machinery—adversarial or equivariant—beats not fighting the gauge, and the only objectives that do no harm are the ones that leave the task alone. The practical payoff is that this is checkable in advance: the continuity of the gauge is a property of the metadata, the covariance of the utility is a property of the task definition, and the recoverability of the group action is testable with a transport-versus-shuffle control before committing to a full training run.

9 Reproducibility

The release is self-contained. A single portable trainer implements all four objectives behind configuration flags—the baseline, the gradient-reversal adversary (single-embedding and dual content/style path), the auxiliary $SO(2)$ equivariance head, and the primary gauge-aligned contrastive objective with both common-reference and pairwise-transport evaluation. The projection-angle dataset adapter reads the CardioSYNTAX shards and exposes the raw positioner angle as the continuous gauge parameter; the pre-registered analyzer emits every confidence interval, permutation p-value, and shuffled-gauge control in this paper. Each reported number is one analyzer verdict over five seeds. The external imaging data are licensed and are not redistributed; the released artifact is the code required to rerun the experiments on the data and the complete per-seed evidence behind every table.

References

Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor Lempitsky. Domain-adversarial training of neural networks, 2016. URL https://arxiv.org/abs/1505.07818.